Maxmilian Duarte

Python Backend Engineer

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PROFESSIONAL SUMMARY

Python backend engineer with 4 years of production experience building REST APIs, async processing systems with Celery and RabbitMQ, and data workflows on advanced SQL. I own systems end to end: design, implementation, tests, and deployment with Docker on Linux.

SKILLS

Backend	Python, Flask, FastAPI, Django, REST APIs, Celery, RabbitMQ, OAuth2 / PKCE, Server-Sent Events
Data	MySQL, SQLite, MongoDB, advanced SQL, stored procedures, ETL
Infra	Docker / Compose, Linux, systemd, nginx, Proxmox / LXC, GitHub Actions, Uptime Kuma, scheduled backups
Quality	pytest, end-to-end tests, type hints, production support (L3), log analysis
Also	TypeScript / Next.js, Vue 3, Kotlin / Jetpack Compose, Selenium / Playwright, FFmpeg

EXPERIENCE

Independent Developer — Jul 2026 – Present
Own products and open source — Remote

- Building MaxTools Radio and cnabkit, an open-source Python library for Brazilian bank files (CNAB 240/400, OFX), on self-managed infrastructure: Proxmox, Docker, nginx, Uptime Kuma, and weekly backups.

Software Engineer (Python, Backend and Data) — Sep 2024 – May 2026
Netbiis | Data Driven Technologies — Remote

- Built and owned an internal Flask and SQL web application for ~32 daily campaign jobs: automated QA fully covered 10 to 15 of them, and its support tools raised operator output by up to 40% across all 32.
- Wrote and optimized complex SQL and stored procedures for high-volume extraction and audience segmentation in Adobe Campaign.
- Provided L3 support for production automation pipelines: log-based root-cause analysis, fixes, and follow-up.

Backend Developer, HG7 and ExportSul (insurance brokerages) — Feb 2024 – Aug 2024
Independent Consultant — Hybrid

- HG7: Built an OFX bank-statement import and reconciliation system (Python, Flask, MySQL) across three iterations, embedded in the company's PHP ERP and feeding its income statement, replacing line-by-line spreadsheet reconciliation.
- HG7: Integrated RD Station CRM and 3CX telephony with internal systems; built ETL pipelines from RD Station and Microsoft Graph into MongoDB for Power BI dashboards.
- ExportSul: Built Selenium bots processing invoices in an ERP, with retries and per-item state so runs resume where they stopped.

Python Developer (Internship) — Aug 2022 – Feb 2024
Plug Benefícios — São Paulo

- Built and maintained 15 production bots (Python, Selenium, Playwright) processing beneficiary changes across 4 health-insurer portals from a MySQL ticket queue, plus invoices, slips, and payslips.
- Designed a Flask + Celery + RabbitMQ service that runs the bots in parallel behind a REST API, with concurrency sized by a load test.
- Extracted shared automation code into an internal Python package (wheel) reused by every bot; used MySQL for transactional data, MongoDB for scraped data, and built internal dashboards in Django and Flask.

Tech Instructor — Oct 2021 – Feb 2022
Microcamp Tecnologia — São Paulo

- Taught programming logic, algorithms, and Python to 50+ students.

PROJECTS · case studies at maxmilian02.github.io

MaxTools Radio: Python bot streaming 48 kHz PCM through a real-time DSP chain, Flask API with SSE and OAuth accounts (Google, Discord, Spotify), web app, and Kotlin / Jetpack Compose Android app. 1,180 Python tests and 340 Android unit tests.

MaxTools: FastAPI plugin platform: a registry auto-discovers tool modules and a Next.js dashboard renders from their manifest. Async jobs with progress, API-key auth, TTL cleanup, and frame-accurate video trimming over HTTP Range.

LANGUAGES

Portuguese (native) · English (advanced: reading, writing, technical documentation)

EDUCATION

B.Sc. Computer Engineering, Impacta Tecnologia (expected 2027)